

SOFTWARE ENGINEERING

LAB-3:Component Modelling & Architectural Pattern Selection

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SECTION:F

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System: Community Lost & Found Matching Network

Architecture Selected: Layered Architecture

Architecture Selection

I chose Layered Architecture for the Community Lost & Found Matching Network.

Architectural Choice:

The system is organized into Presentation, Business, and Data layers. The Presentation Layer handles user interactions, the Business Layer handles report management, matching, claim processing, and administration, while the Data Layer manages persistent system information.

Reason 1 – Separation of Concerns:

The system contains different responsibilities such as user interaction, report management, item matching, claim processing, administration, and data storage. Layered Architecture separates these responsibilities into appropriate layers, making the system easier to understand and maintain.

Reason 2 – Maintainability:

Major system functions can be implemented as separate business components such as Report Management, Matching Service, Claim Management, and Admin Management. Changes to one component can therefore be made with minimal impact on other parts of the system.

Security Advantage:

The Business Layer can control access to sensitive claim information before it is retrieved or modified. This helps restrict private claim details to authorized users and Community Admins.

Performance Benefit:

The Matching Service can be optimized separately to efficiently search and compare lost and found reports using item details, location, and date. This supports the requirement that potential matches should be displayed within five seconds after report submission.

UML Component Diagram for Community Lost & Found Matching Network

